CHAPTER

1

MEng 2131
Thermodynamics I

Introductory concepts and definitions

Thermodynamics Structure

- The behaviors of the system can be studied from:-

1. **Macroscopic point of view**
   - The macroscopic approach to thermodynamics is concerned with the **gross** or **overall behavior**.
   - This is some times called **classical thermodynamics**
   - No model of the structure of matter at the molecular, atomic and subatomic levels is directly used in classical thermodynamics.

Thermodynamics Structure

2. **Microscopic point of view**
   - concerned with the **structure of matter**.
   - This is some times called **statistical thermodynamics**
   - Average behavior of the particles making up a system.

- For the great majority of engineering application, classical thermodynamics not only provides a considerably more ***direct approach for analysis and design*** but also requires far fewer **mathematical application**.

Thermodynamics System

- An important step in any engineering analysis is to describe precisely **what is being studied**.
- In mechanics, if the motion of a body is to be determined normally:-
  - the first step is to define a free body diagram.
  - Identifying all the forces exerted on it by other bodies
  - Newton's second law of motion is then applied

Thermodynamics System

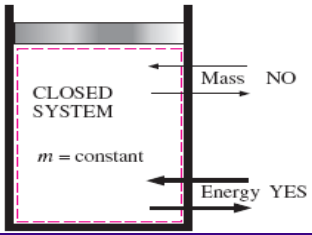
- In thermodynamics :-
  - the first step is to define a system.
  - Once the system is defined and the relative interaction with the other systems are identified.
  - One or more physical laws are applied

Closed, Open and Isolated System

- In thermodynamics ***system*** is defined as a quantity of matter or a region in space chosen for study.
- The region outside the system is called the ***surroundings***.
- The real or imaginary surface that separates the system from its surroundings is called the ***boundary***.
- The boundary of a system can be ***fixed or movable.***
- Systems may be considered to be ***closed or open,*** depending on whether a fixed mass or a fixed volume in space is chosen for study.

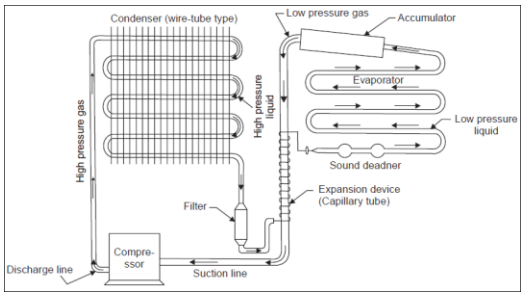
Closed System

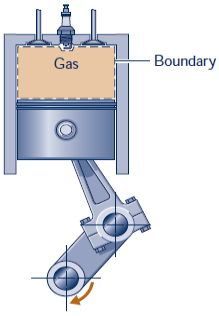
- Also called ***non-flow*** system.
- Consists of a fixed amount of mass and ***no mass*** may cross the system.
- The closed system boundary may move.
- Energy in the form of ***heat and work*** can cross the boundaries of a closed system.

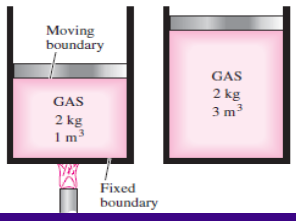


Example

- Sealed tanks
- Piston cylinder device
- Refrigeration system

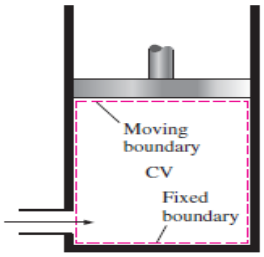






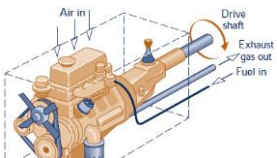
Open System

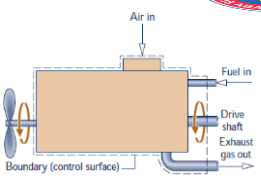
- Also called ***flow*** system.
- ***Mass*** as well as ***energy*** crossing the boundary.
- It is called a ***control surface*** because both mass and energy can cross the boundary of a ***control volume***.



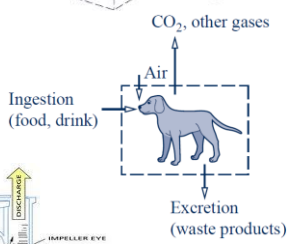
Example

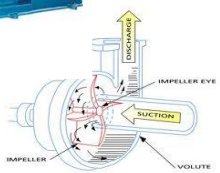
- Pumps
- Compressor
- Heat exchanger
- Engines

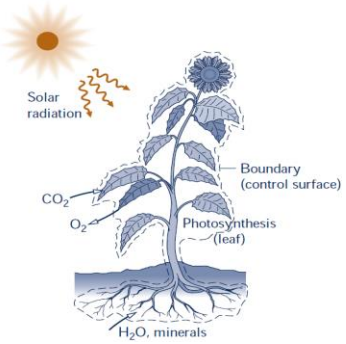






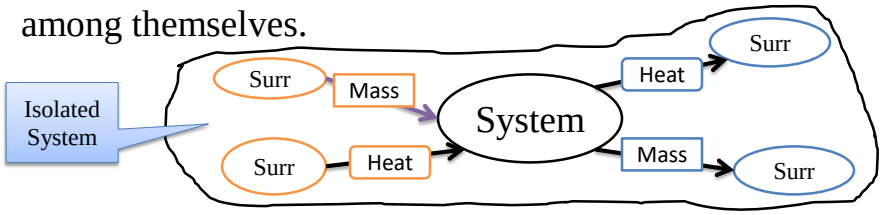






Isolated System

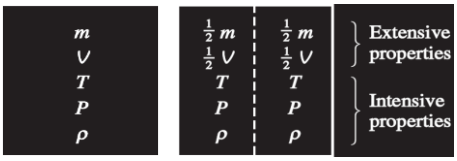
- A general system of fixed mass where ***no heat*** or ***work*** may cross the boundary.
- A closed system with no energy crossing the boundaries and is normally a collection of a main system and its surrounding that are exchanging mass and energy among themselves.



State, Equilibrium, Process and Properties

Property

- Any characteristic of a system is called a ***property***.
- Some familiar properties are ***pressure*** *(P)*, ***temperature*** *(T)*, ***volume*** *(V)*, *and* ***mass*** *(m)*.
- There are two types of property; ***intensive*** *or* ***extensive***
  - ***Intensive*** **properties** -are those that are independent of the ***size*** of a system.
    - Temperature,
    - Pressure
    - Age
    - Color
  - ***Extensive*** **properties** -are those that are depend on the size of the system.
    - Mass
    - Volume
    - Total energy

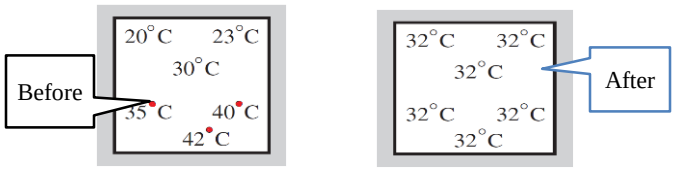


State

- Condition of the system at an instant of time as described or measured by its properties.
- Consider a system not undergoing any change. The properties can be measured or calculated throughout the entire system. This gives us a set of properties that completely describes the condition of a state.
- At a given state, all the properties are known; changing one property changes the state.

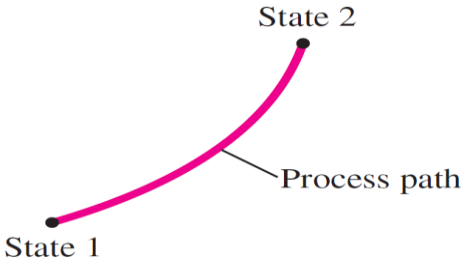
Equilibrium

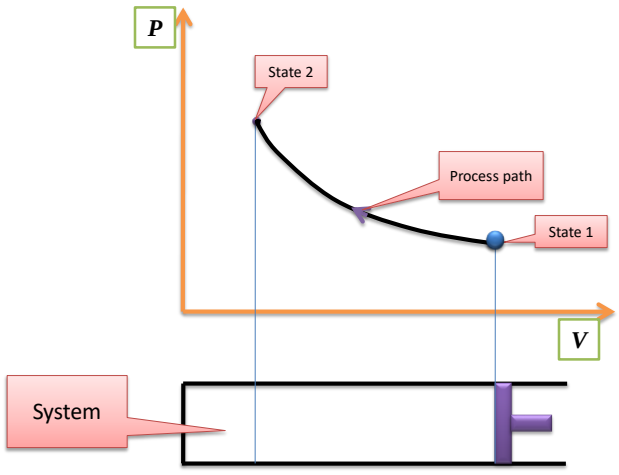
- The word ***equilibrium*** implies a state of balance.
- A system is said to be in thermodynamic equilibrium if it maintains thermal (uniform temperature), mechanical (uniform pressure), phase (the mass of two phases, e.g ice and liquid water, in equilibrium) and chemical equilibrium.



Process

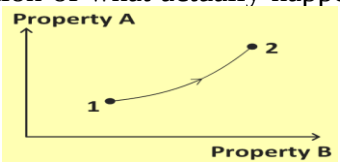
- When any of the properties of a system change, the state changes and the system is said to have undergone a ***process.***
- A ***process*** is a transformation from one state to another.





Quasi-static process

- A quasi-static process is one :-
  - ✓occurs *sufficiently slowly.*
  - ✓The deviation from thermodynamic equilibrium is *infinitesimal.*
  - ✓All states of the system passes through are equilibrium states.
- A quasi-static process is an idealization of what actually happens, but it is a reasonable approximation.



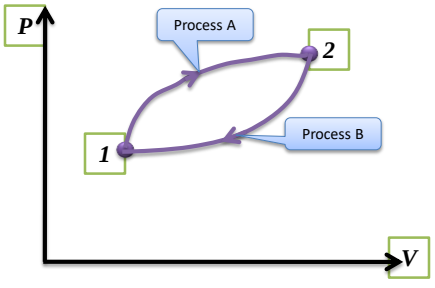
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- In most of the process that we will study, one thermodynamic properties is held ***constant***.
- Some of the process are:

Process	Property held constant
Isobaric	Pressure
Isothermal	Temperature
Isochoric	Volume
Isentropic	Entropy

Cycle

- Series of process in which the ***initial state*** of the first process is ***identical*** to the ***final state*** of the last process.



Dimensions and Units

- Any physical quantity can be characterized by ***dimensions***.
- The arbitrary magnitudes assigned to the dimensions are called ***units***.
- In engineering thermodynamics, basically, there are ***four*** dimensions. (primary dimension)
  - Mass, Length, Time and Temperature
- The others are simply derived from primary units.

Some basic properties of thermodynamics



Specific Volume

- The volume occupied by a unit mass of a substance, and it is designated by **n.**

$$v = \frac{Volume}{mass} = \frac{V}{m}\ (m^3/kg) = \frac{1}{\rho}$$

- Specific density /gravity
  - The density of substance divided by the density of water.

$$\rho_s = \frac{\rho}{\rho_{H_2O}}, \rho_{H_2O} = 1000 kg/m^3$$



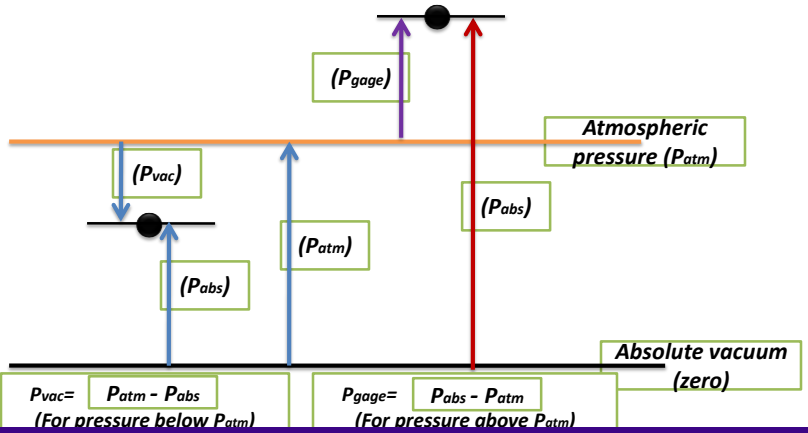
Pressure

- Pressure is defined as *a normal force exerted by a fluid per unit area.*

$$P = \frac{Force}{Area} = \frac{F}{A} = \frac{N}{m^2} = Pascal = Pa$$

- The actual pressure at a given point is called the ***absolute pressure*** and it is measured relative to ***absolute vacuumed***.(absolute ***zero pressure***)
- However, pressure are often measured ***relative to atmospheric*** pressure, called ***gage*** or ***vacuum*** pressure.

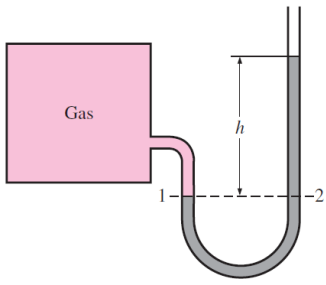
Atmospheric pressure is the force per unit area exerted into a surface sea level by the weight of air above that surface in the atmosphere of earth.



Manometer

- Measure small and moderate pressure differences.
- Manometer mainly consists of a glass or plastic ***U-tube*** containing one or more fluids such as mercury, water, alcohol, or oil.
  - Pressure is the same in the horizontal direction for a fluid at rest

$$P_2 = P_{\text{atm}} + \rho g h$$



Temperature

- It is considered as a thermodynamic property that is the measure of the ***energy content*** of a mass.
- When ***heat*** energy is transferred to a body, the body's ***energy*** content increases and so does its ***temperature***.
- Two bodies are in ***thermal equilibrium*** when they have reached the ***same temperature***.
- If two bodies are in ***thermal equilibrium*** with a ***third body***, they are also in thermal equilibrium with ***each other***. This simple fact is known as the ***zeroth law of thermodynamics.***

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- If two objects are in thermal equilibrium with a third object, then they are in thermal equilibrium with each other.

Thermal equilibrium => same temperature

- The Zeroth Law gives a formal underpinning (foundation) of the *thermometer concept*.
- The thermometer is the third object that tells us the temperature of the objects are the same.

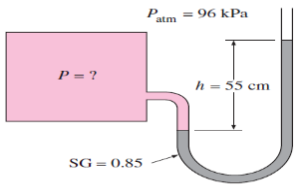
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Examples

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1. A manometer is used to measure the pressure in a tank. The fluid used has a specific gravity of 0.85, and the manometer column height is 55 cm, as shown in Fig. If the local atmospheric pressure is 96 kPa, determine the absolute pressure within the tank.

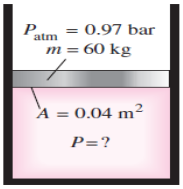


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2.The piston of a vertical piston–cylinder device containing a gas has a mass of 60 kg and a cross-sectional area of 0.04 m^2, as shown in Fig. The local atmospheric pressure is 0.97 bar, and the gravitational acceleration is 9.81 m/s^2.

(a) Determine the pressure inside the cylinder.

(b) If some heat is transferred to the gas and its volume is doubled, do you expect the pressure inside the cylinder to change?



3.A gas is contained in a vertical, frictionless piston–cylinder device. The piston has a mass of 4 kg and a cross-sectional area of 35 cm^2. A compressed spring above the piston exerts a force of 60 N on the piston. If the atmospheric pressure is 95 kPa, determine the pressure inside the cylinder.

